

1 Mixture-path construction

Introduce an auxiliary variable $Z = (X_0, X_1)$ and define, for each coordinate i , the conditional path

$$p_{t|0,1}^i(x_i | x_0, x_1) = \kappa(t) \delta(x_i, x_{1,i}) + (1 - \kappa(t)) \delta(x_i, x_{0,i}),$$

we take for simplicity:

$$\kappa(t) = t, \quad \dot{\kappa}(t) = 1,$$

so that at time t , each coordinate $X_t^i = X_1^i$ with probability t and $X_t^i = X_0^i$ with probability $1 - t$. Under this choice, at time t each coordinate X_t^i equals X_1^i with probability t and X_0^i with probability $1 - t$:

```
t = torch.rand(batch_size) # t ~ Uniform[0,1)
x_t = torch.where(torch.rand(batch_size, 2) < t[:, None], x_1, x_0)
```

2 Kolmogorov forward equation

Recall that for our mixture-path

$$p_{t|0,1}^i(x_i | x_0, x_1) = t \delta(x_i, x_{1,i}) + (1 - t) \delta(x_i, x_{0,i}),$$

we must solve the Kolmogorov forward equation

$$\frac{d}{dt} p_{t|0,1}^i(y_i | x_0, x_1) = \sum_{x_i} u_t^i(y_i, x | x_0, x_1) p_{t|0,1}^i(x_i | x_0, x_1).$$

A direct calculation of the left-hand side gives

$$\frac{d}{dt} p_{t|0,1}^i(y_i | x_0, x_1) = \delta(y_i, x_{1,i}) - [t \delta(y_i, x_{1,i}) + (1 - t) \delta(y_i, x_{0,i})].$$

Matching both sides and using $\kappa(t) = t$, $\dot{\kappa}(t) = 1$ yields the coordinate-wise velocity

$$u_t^i(y_i, x | x_0, x_1) = \frac{1}{1 - t} [\delta(y_i, x_{1,i}) - \delta(y_i, x_i)].$$

One checks immediately that $u_t^i(\cdot, x) \in \mathbb{R}^v$ satisfies the CTMC rate-conditions (nonnegativity off-diagonal and zero row-sum).